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A test of the umbrella species approach in restored floodplain ponds

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Summary

1. The umbrella species approach, where conservation actions targeted for one or a group of species should benefit the broader community, may provide an effective framework to guide habitat restoration. This requires congruence in the response of umbrella and co-occurring species to environmental stress and recovery, and the identification of potential mechanisms by which co-occurring species benefit from conservation of an umbrella species.

2. Past evaluations of this approach have considered only the presence/absence of umbrella species. In addition to the presence/absence, we quantified abundance and biomass of both umbrella and co-occurring species to support a more quantitative evaluation of species co-occurrence. Floodplain ponds are restored in British Columbia and the Pacific Northwest to benefit coho salmon *Oncorhynchus kisutch,* a presumptive umbrella species. To test its effectiveness as an umbrella species, we assessed the relationships between species richness, abundance and biomass of aquatic vertebrates, including vertebrates of conservation concern ('listed'), and benthic invertebrates and the abundance and biomass of juvenile coho. We used ordination to evaluate relationships between species' abundance and biomass and environmental attributes.

3. Positive relationships were identified between coho abundance and biomass and species richness, abundance and biomass of fish and listed species. These relationships were negative for benthic invertebrates. Listed species were located close to coho in ordinations, suggesting they respond to similar environmental features while benthic invertebrates clustered away from coho.

4. *Synthesis and applications*. We found that where our umbrella species coho is most productive, so are other listed species and fish in general, a relationship that would not have been evident had evaluated species richness alone. We also reported strong relationships between some environmental features manipulated in the habitat restoration and the presence and productivity of coho and co-occurring species. Our study demonstrates the umbrella species approach has potential to guide habitat restoration when there is congruence in the response of umbrella and co-occurring species to environmental attributes that can be manipulated in the restoration. Where this is possible, one can use restoration designed for one or several umbrella species and successfully restore habitats that are viable for other species, including listed species.

Key-words: amphibians, coho salmon, fish, habitat restoration, invertebrates, listed species

Introduction

One of the most effective ways to conserve biodiversity, including ecosystems, biological assemblages, species and populations, is to set aside protected areas that encompass a representative sample of biodiversity, maintain natural processes and viable populations, and exclude threats (Margules & Pressey 2000). If the conservation of biodiversity is to grow beyond the limited areas available for preservation and protection in reserves (Vitousek *et al.* 1997), then approaches used to design reserves need to be applied more broadly in degraded terrestrial and aquatic systems with conservation potential (e.g. agricultural, light residential). The umbrella species concept, in which conservation measures designed to benefit one or more species should confer benefits to populations of co-occurring

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species, has been evaluated for its use in reserve design and for its potential to determine the size and/or structural characteristics of an area to be protected or managed (Caro & O'Doherty 1999; Roberge & Angelstam 2004; Branton & Richardson 2011). The need to validate the effectiveness of potential umbrella species in providing benefits to co-occurring species has long been recognized (Caro & O'Doherty 1999). Some researchers have also advocated the identification of mechanisms by which co-occurring species benefit from conservation of an umbrella species, such as specific resource requirements or dispersal corridors (e.g. Lambeck 1997; Ozaki *et al.* 2006). The application of the umbrella species concept holds potential for restoration planning, particularly if presumed umbrella and co-occurring species respond similarly to restoration.

Species richness is the most common measure of the effectiveness of potential umbrella species despite the fact the umbrella species approach was originally focused on population sizes and viability (i.e. population unlikely to go into rapid decline, Caro 2003) of both umbrella and co-occurring species (e.g. Berger 1997). This is problematic from a conservation planning perspective because species richness provides no information on species density or demography, and therefore about persistence of populations, or about the composition of ecological communities (Fleishman, Noss & Noon 2006). Umbrella species may be more valuable for conservation planning if both their presence and abundance vary similarly to co-occurring species in response to disturbance or restoration (Fleishman, Murphy & Brussard 2000; Koper & Schmiegelow 2007). When this is the case and the spatial transferability of one or more umbrella species across a region with common ecology and ecological challenges can be established, the design and monitoring of conservation and restoration planning can be facilitated across that region (e.g. Pakkala, Pellikka & Linden 2003; Betrus, Fleishman & Blair 2005).

In the Pacific coastal ecoregion of North America, off-channel floodplains, including sloughs, side channels and beaver ponds, provide rearing and overwintering habitat for juvenile salmonids such as coho salmon *Oncorhynchus kisutch* (Walbaum, 1792) (hereafter 'coho') (Sandercock 1991). The channelization of rivers and isolation of floodplains have simplified and reduced these habitats, the scarcity of which may limit coho production (Beechie, Beamer & Wasserman 1994). We used data collected from floodplain ponds that have been restored, created and enhanced (hereafter 'restored') primarily for coho in south-western British Columbia, Canada to evaluate whether other aquatic vertebrate and benthic invertebrate species, including species of conservation concern (listed species), benefitted from those restoration projects, as measured by species richness, abundance and biomass. Given the imperilled status of coho and other Pacific salmon (Pacific Fisheries Resource Conservation Council 2010), regional conservation programmes, including habitat restoration, are likely to continue into the future. In this context, an iterative approach that uses data collected from monitoring and assessment of restoration projects to provide insights for the modification of future projects that may benefit coho and co-occurring species is imperative (Roni 2005).

We tested whether species richness, abundance and biomass of co-occurring species varied in a pattern similar to the response variables related to the population viability (*sensu* Caro 2003) of coho (i.e. abundance and biomass per unit effort). We also evaluated whether variation in species' abundance and biomass could be explained by environmental attributes of the restored ponds. We tested the generality of the results by sampling in multiple ponds in three watersheds within the same region. Although species composition, abundance and biomass of the taxonomic groups evaluated may vary from year to year, we restricted the sampling to 1 year to represent one cohort of coho to avoid confounding factors associated with interannual variability in coho populations that can be high due to many factors including oceanic conditions and fisheries pressure.

Materials and methods

STUDY SITES

Aquatic systems in the Fraser River Basin of south-western British Columbia, Canada, (Fig. 1) have been impaired by impacts including the loss of riparian vegetation, water diversion and stream channelization. All 100 off-channel restoration projects implemented by Fisheries and Oceans Canada (DFO) as of 2004, primarily to benefit juvenile coho in south-western British Columbia, were candidate study sites (Pers. Comm., Matt Foy, DFO). Ponds were restored by improving the connectivity of existing ponds to surface water and by creating new ponds using groundwater or water diverted from nearby dams, rivers and creeks to flood areas within a berm or in excavated areas. Wood features were commonly added and deep channels created as part of the restoration. Restoration techniques used and physical features of each pond are summarized in Table S1 in Supporting information.

The criteria used to select sites for inclusion in our study included the presence of ponds, no direct tidal influence, surface or groundwater fed (not glacial), accessibility, no stocking of fish (including coho) and a minimum of four restored ponds per watershed. Seventeen restored floodplain ponds in three watersheds met our criteria: Chilliwack (n = 9), Coquitlam (n = 4) and Seymour (n = 4) (Fig. 1).

FIELD SAMPLING

Vertebrate sampling

We sampled ponds three times: (i) late spring (May–June) 2006, (ii) summer (July–August) 2006 and (iii) winter (February–March) 2007, prior to freshets that would initiate the outmigration of coho smolts. We selected these study periods to increase the likelihood of detecting species present in ponds only for cer-

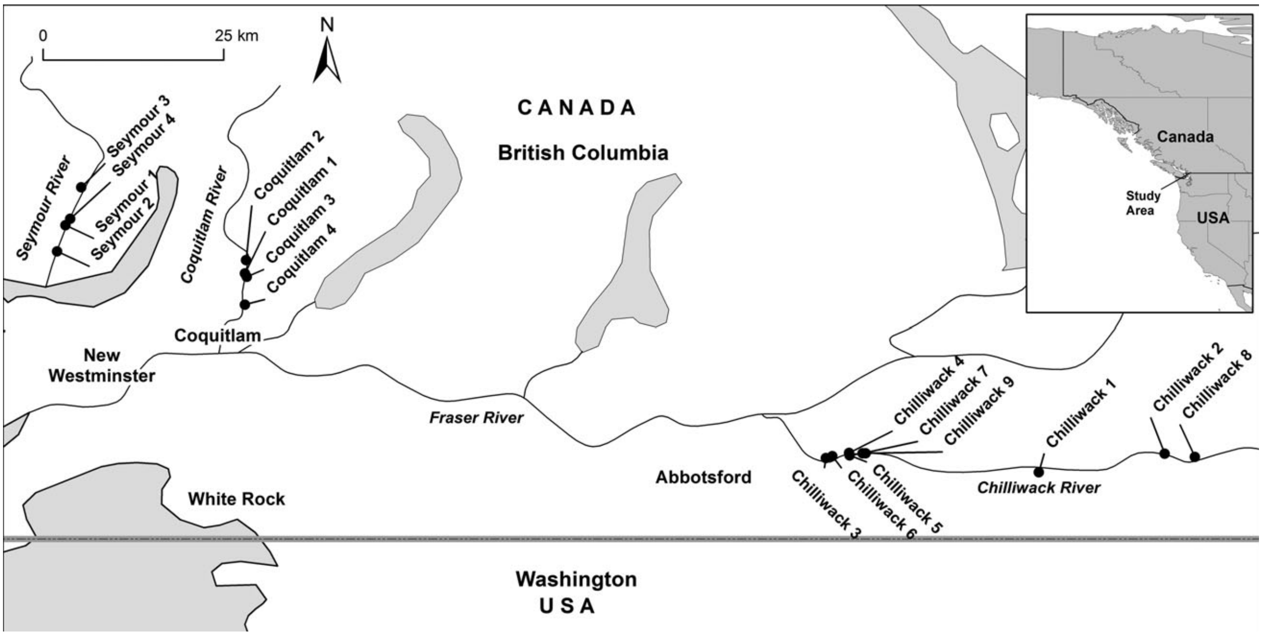

Fig. 1. Location of study sites in south-western British Columbia, Canada. The study sites have a northerly midpoint of *c.* 49 16′ N and extend from 121° 30′ W to 123° 4′ W.

tain life stages (i.e. metamorphosing amphibians). For coho, this sampling period was intended to capture a cohort exposed to similar environmental conditions both in the marine environment for adult spawners and in freshwater for their offspring (the cohort of interest).

Between 30 and 50 minnow traps, depending on pond size, baited with salmon roe were set in each pond. A visual assessment was used to divide each pond into sections using features such as depth, aspect, riparian structure and aquatic structure to ensure sampling of all representative habitat types (Olson, Leonard & Bury 1997). Approximately, equal numbers of traps were set haphazardly in each section of the pond. Captured juvenile and adult fish and amphibians were counted, identified to species level, weighed and fork length (fish) or snout-vent and total length (amphibians) were measured (Barbour *et al.* 1999; Corkran & Thoms 2006). We anaesthetized fish using buffered MS222 prior to taking fork length and mass measurements. Approval from the University of British Columbia Animal Care Committee and federal and provincial permits were obtained.

Benthic invertebrate sampling

In late spring 2006 (May–June), we collected three replicate benthic invertebrate samples from each pond using a standard D-frame to collect semiquantitative travelling kick net samples, standardized by time (Wissinger, Greig & McIntosh 2009). Sampling locations were selected using a random number table or by access combined with aspect if there were access limitations (e.g. deep mud, conglomerations of wood). Samples were preserved in 10% formalin in the field, sorted in the laboratory, counted and identified to the lowest practical level (family or genus). Large samples were subsampled. For biomass estimates, all individual benthic invertebrates from each pond were aggregated, irrespective of species, blotted dry and weighed to the nearest 0·01 g.

Habitat assessment

We documented each pond's habitat structure in summer 2006 by surveying the presence or absence of wood, macrophytes, algae and benthic organic matter, percentage overhead riparian cover and water depth every metre along four to six equidistant transects of each pond (determined by the size of the pond) (Anonymous 1995; Johnston & Slaney 1996). The proportion of all measurements for each structural component was calculated (e.g. 30 out of a total of 100 readings = 0·3). All ponds had predominantly fine substrate except in small areas with gravels at stream mouths. We placed temperature loggers at two depths (*c.* 30 and 100 cm below the water surface) in each pond from May or June 2006 to July 2007. Due to the loss or malfunction of loggers, particularly from August 2006 to February 2007, we relied exclusively on maximum temperatures calculated as the average temperature of the warmest 7-day period. We estimated pond area using the area estimation function of Google Earth (Version 3.0, Google Inc., Mountain View, CA, USA).

Analyses

We evaluated relationships between species richness, abundance and biomass of co-occurring species and that of coho abundance and biomass, which varied widely across ponds. For all analyses, we categorized species by broad taxonomic groupings as follows: fish excluding coho ($n = 13$); fish excluding coho and threespine stickleback *Gasterosteus aculeatus* (Linnaeus, 1758) ($n = 12$); amphibians ($n = 6$); and all benthic invertebrates identified to the lowest practical level ($n = 60$). A summary of species by category and full species lists are provided in Table S2 (Supporting information). Vertebrates were categorized by taxonomic group to permit the evaluation of the suitability of coho as an umbrella species within and across taxa. Fish were tested with and without threespine stickleback as the relationship between coho, and other fish species would have been obscured by the relationship between coho and threespine stickleback alone. Threespine stickleback were extremely abundant comprising an average of 97% (SD = 3) of total fish abundance and 69% (SD = 30) of biomass (excluding coho) in eight of the nine ponds where they occurred. In the ninth pond, threespine stickleback accounted for only 3% of abundance and 0·1% of biomass of fish, and in the remaining eight ponds, no threespine stickleback was documented.

We used a final category for fish and amphibians listed as 'species at risk' or 'of special concern' (listed) by federal or provincial conservation authorities. The northern red-legged frog *Rana aurora* (Baird and Girard, 1852) is both federally and provincially listed and cutthroat trout *O. clarkii* (J. Richardson, 1836) and Dolly Varden *Salvelinus malma* (Walbaum, 1792) are listed provincially. The Salish sucker *Catostomus* sp. *cf. catostomus* (Lesueur, 1817) has an endangered designation, which reflects its low population numbers and very limited global distribution, and is the subject of its own recovery plan (Recovery Team for Salish Sucker, 2010). The Salish sucker was therefore excluded from the listed category as the umbrella species approach is intended to provide conservation benefit for species that are equally or less sensitive than the umbrella species and that have viable populations (Caro 2010).

Abundance and biomass

We calculated abundance and biomass per unit effort (i.e. per trap-night) by dividing the total number or biomass of individuals captured in a pond by the total number of traps used in that pond in a sampling period. In some sampling periods, a large number of individuals of some species [e.g. threespine stickleback, northwestern salamander *Ambystoma gracile* (Baird, 1859)] were captured. In situations where more than 15 individuals of one species were captured, we measured the first five individuals in each subsequent sampling section and additional individuals were assigned to length classes (e.g. 4–5 cm). We estimated biomass for nonweighed individuals by assigning them the mean biomass for conspecific individuals from their length class in that sampling period. Fish and amphibians that escaped prior to measurement were assigned to length categories, and their biomass was estimated.

Species richness

We used sample-based rarefaction curves with 1000 iterations, without replacement, in a Monte Carlo type analysis, to calculate rarefied species richness using the expected richness function (Mau Tau) in EstimateS (Colwell 2009). The pond with the fewest individuals in a category that was greater than zero was used to determine how many individuals species richness was rarefied to. When this would have resulted in rarefying to an extremely small number of individuals (e.g. three or less), we rarefied to the lowest value five or greater (arbitrary cut-off). Ponds with a species richness of zero for a category were retained in the analysis based on the assumption the zero value represented the real absence of species in that category from the pond.

Regressions and multivariate analyses

We evaluated relationships between coho productivity and the richness and productivity of co-occurring species in restored ponds using the random and repeated functions in mixed models (PROC MIXED) (SAS 9.1, SAS Institute, Cary, NC, USA). Watershed was the random variable; coho abundance and biomass were the fixed explanatory variables; and rarefied species richness, abundance and biomass were the dependent response variables. The three sampling sessions were the repeated measures. PROC GLIMMIX was used for listed species only as the data did not meet the assumptions of normality. Both watershed and sampling sessions were treated as random variables in this model. When there was zero variance associated with watershed or sampling session, no random variable was used. A pseudo-R-squared was calculated for analyses conducted with PROC MIXED using sums of squares (1-SSE/SSE).

We transformed data to normalize residuals and meet the assumption of normality for general linear models. For dependent variable species richness, the independent (coho abundance and biomass) variables were square-root-transformed for listed species and both dependent and independent variables were $\log_e$-transformed for fish. For dependent variables, abundance and biomass, both dependent and independent (coho abundance and biomass) variables were $\log_e$-transformed for listed species and invertebrates; both dependent and independent variables were square-root-transformed for fish (no coho) and for fish (no coho, no stickleback); and the independent variables alone were $\log_e$-transformed for amphibians. Tests were considered significant at alpha = 0·1 to reduce the likelihood of a type II error, which would be more likely with alpha = 0·05 because of large sampling variability or small sample size (Peterman 1990; Bryant 2004).

We used redundancy analysis to evaluate the relationship between species abundance and biomass and environmental variables (ter Braak & Smilauer 2002) (CANOCO 4.5). Ordinations were used to illustrate the distribution of vertebrate and benthic invertebrate species abundance and biomass along the first two environmental axes. Benthic invertebrate species that were not present in at least three ponds were excluded from the analysis. Data were not transformed to meet the assumptions of normality as the ordination uses a Monte Carlo analysis that does not assume a normal distribution; however, the original species data included many zeros so we used a log (x + 1) transformation for species data (ter Braak & Smilauer 2002). We scaled species data by dividing species scores by their standard deviation and standardized the data using species error variance to counteract rare species unduly dominating ordination. The percentage of the total variance (i.e. inertia) in the species data explained by environmental variables is given by the species–environment relationship. We used a global Monte Carlo permutation test (499 permutations) to determine the statistical significance of the relationship between the species and environmental variables represented by the first canonical axis alone and for all four axes together. Correlations between each axis and environmental variables were considered significant ($P < 0{\cdot}05$) at a critical value of $r = 0{\cdot}48$.

Results

SPECIES RICHNESS

There were significant positive relationships between the species richness of listed species and fish and coho abundance and between the species richness of fish and coho biomass (Table 1, Fig. 2a,b). Benthic invertebrate species richness decreased significantly as coho abundance and

Table 1. Analysis of the relationships between dependent variables species richness, abundance and biomass of co-occurring species of conservation concern (listed species), fish, amphibians, benthic invertebrates and the abundance and biomass of coho salmon, the presumed umbrella species. Significant results at Alpha = 0·1 in bold

	Coho abundance[¶]	Pseudo R^2	Coho biomass	Pseudo R^2
Species richness				
Listed species[†]	**$F_{1,49}$ = 3·60, P = 0·06**	na	$F_{1,47}$ = 1·20, P = 0·28	na
Fish (no Coho)*	**$F_{1,47}$ = 9·86, P < 0·01**	0·26	**$F_{1,47}$ = 4·86, P = 0·03**	0·16
Amphibians	$F_{1,33}$ = 0·00, P = 0·98	0·00	$F_{1,33}$ = 0·05, P = 0·83	0·00
Invertebrates	**$F_{1,47}$ = 8·43, $P \leq$ 0·01**	0·39	**$F_{1,47}$ = 8·43, $P \leq$ 0·01**	0·23
Abundance or biomass**				
Listed species[§]	**$F_{1,33}$ = 4·03, $P \leq$ 0·01**	na	**$F_{1,33}$ = 85·42, P < 0·01**	na
Fish (no Coho)*	**$F_{1,33}$ = 25·44, P < 0·01**	0·33	$F_{1,33}$ = 2·19, P = 0·15	0·04
Fish (no Coho, no stickleback)*	**$F_{1,33}$ = 6·96, P = 0·01**	0·12	**$F_{1,47}$ = 16·49, $P \leq$ 0·01**	0·30
Amphibians[‡]	$F_{1,33}$ = 0·57, P = 0·46	0·01	$F_{1,33}$ = 0·65, P = 0·43	0·01
Invertebrates[§]	**$F_{1,33}$ = 13·4, $P \leq$ 0·01**	0·21	**$F_{1,33}$ = 12·56, P < 0·01**	0·20

Transformations
*Square-root-dependent and independent variable.
[†]Square-root-independent variable.
[‡]Log$_e$-independent variable.
[§]Log$_e$-dependent and independent variable.
[¶]PROC GLIMMIX used for listed species only. PROC MIXED used for all other analyses. df = 47 with random variable using PROC MIXED; df = 33 without random variable using PROC MIXED; df = 49 using PROC GLIMMIX with season as random variable.
**Response variables biomass and abundance with, respectively, explanatory variables biomass and abundance.

biomass increased (Table 1, Fig. 2c). There were no significant relationships between amphibian species richness and coho abundance or biomass (Table 1).

ABUNDANCE AND BIOMASS

The abundance and biomass of listed species increased as those same measures increased for coho (Table 1, Fig. 2d). There was a significant negative relationship between the abundance of fish and that of coho, however, when threespine stickleback were excluded from the model that relationship became positive (Table 1, Fig. 2e,f). Similarly, there was no significant relationship between the biomass of fish and coho, however, when threespine stickleback were excluded from the model, that relationship was positive (Table 1). As coho abundance and biomass increased, benthic invertebrate abundance and biomass decreased (Table 1, Fig. 2g). There were no significant relationships between amphibian abundance or biomass and that of coho (Table 1).

VARIANCE IN ABUNDANCE AND BIOMASS EXPLAINED BY ENVIRONMENTAL GRADIENTS

Environmental attributes explained 30·3% of the variance in aquatic vertebrate species (including coho) abundance data using axis 1 alone, and the cumulative variance explained by axes 1 and 2 was 48·4%. The relation between species and environment was significant for the first axis (eigenvalue = 0·27, P = 0·01) and for all canonical axes (Trace = 0·88, P = 0·002). Aquatic vegetation, elevation, maximum temperature and wood were correlated with axis 1 (Fig. 3a); organic matter was correlated with axis 2; depth at water's edge was correlated with axis 3; and variation in depth was correlated with axis 4. In the ordination, salmonids, including coho, were clustered on the negative side of axis 1, which was characterized by higher elevation, more wood, groundwater inputs, lower maximum temperatures and less aquatic vegetation. Northern red-legged frog was also on the negative portion of axis 1 but, such as chinook salmon *O. tshawytscha* (Walbaum, 1792), the relationship with the environmental variables, as indicated by the length of the arrows in the direction of the significant environmental feature, was not as strong.

Environmental variables explained 28·7% of the variance in aquatic vertebrate species biomass on axis 1 and the cumulative variance explained by axes 1 and 2 was 47·5%. The relationship between species and environment was significant for the first (eigenvalue = 0·25, P = 0·04) and for all canonical axes (Trace = 0·88, P = 0·002). Aquatic vegetation, elevation, maximum temperature, water source and wood were significantly correlated with axis 1. Organic matter and maximum depth were correlated, respectively, with axes 2 and 4. No variables were correlated with axis 3 (Fig. 3b). Salmonids (including coho), northern red-legged frog and northern Pacific treefrog *Pseudacris regilla* (Baird and Girard, 1852) were clustered on the negative portion of axis 1, which was characterized as higher elevation, more wood, less aquatic vegetation, groundwater inputs and lower maximum temperatures. On axis 2, minnows *Cyprinidae* spp., longnose dace *Rhinichthys cataractae* (Valenciennes, 1842) and north-western salamander clustered where there was more organic matter and away from the remainder of the species.

Environmental variables explained 21·8% of the variance in benthic invertebrate species and coho abundance on axis 1 and the cumulative variance explained by axes 1 and 2 was 41·0%. The relationship between species and

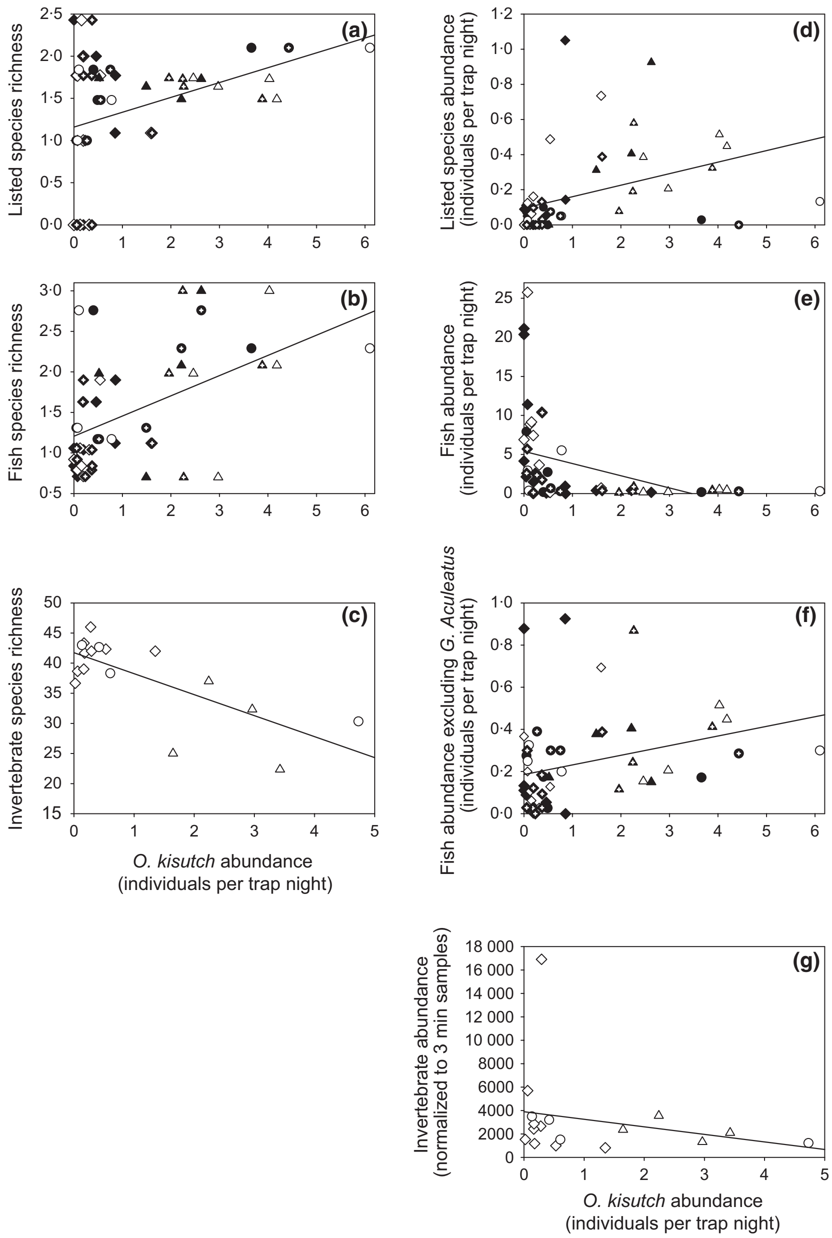

Fig. 2. Relationships between relative abundance of presumed umbrella species coho and species richness of (a) listed species, (b) fish and (c) benthic invertebrates and abundance of (d) listed species, (e) fish, (f) fish excluding threespine stickleback and (g) benthic invertebrates. Average coho abundance over three sampling sessions used for relations with benthic invertebrates (panels c and g). Watersheds distinguished for each relation (◇-Chilliwack, ○-Coquitlam, △-Seymour) and sampling sessions shaded to distinguish sampling sessions one through three (respectively shaded, open and shaded with white cross).

environment was significant for the first canonical axis (eigenvalue = 0·18, $P = 0{\cdot}02$) and for all canonical axes combined (Trace = 0·84, $P = 0{\cdot}002$). Aquatic vegetation, elevation, maximum temperature, organic matter, variation in depth and wood were correlated with axis 1. Algae, variation in depth and area were correlated, respectively, with axes 2, 3 and 4. In general, few benthic invertebrates were clustered near coho (Fig. 4a). Relatively more benthic invertebrates were on the positive side of axis 1, which was characterized by lower elevations, warmer maximum water temperatures, more aquatic vegetation, less wood and organic matter and a more uniform depth profile. While algae was strongly positively correlated with axis 2, with more algae present closer to 1·0 on the axis, benthic invertebrates were equally distributed along the axis. Coho was located around the zero mark of axis 2 indicating no relationship with algae.

Environmental variables explained 74·1% of the variance in biomass of benthic invertebrate species and coho on axis 1 and the cumulative variance explained by axes 1 and 2 was 100%. The high variance explained was due to the fact only two independent constraints could be formed with the environmental variables. The relationship between species and environment was significant for the first (eigenvalue = 0·71, $P = 0{\cdot}01$) and for all canonical axes (Trace = 0·96, $P = 0{\cdot}006$). Aquatic vegetation, maximum temperature, variation in depth and wood were correlated with axis 1, and area was correlated with axis 2 (Fig. 4b). Benthic invertebrates and coho were located on opposite sides of axis 1 with more benthic invertebrate biomass in ponds characterized by more aquatic vegetation, less wood, higher maximum temperatures and less variation in depth. Benthic invertebrates and coho were at approximately the same level on axis 2 indicating a

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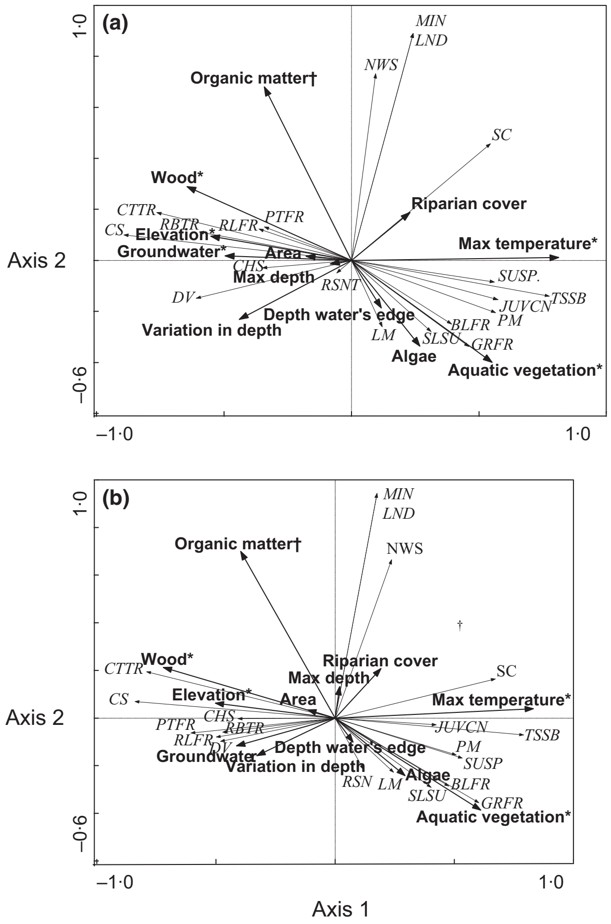

Fig. 3. Redundancy analysis ordination of relative abundance (a) and biomass (b) of vertebrate species (small arrowheads) and environmental attributes (large arrowheads). Significant correlations marked, *(Axis 1) and †(Axis 2). Axis 1: elevation, groundwater and wood increase towards −1·0 and maximum temperature and aquatic vegetation increase towards 1·0. Axis 2: organic matter increases towards 1·0 and aquatic vegetation increases towards −0·8. Fish: CHS chinook salmon, CS coho salmon, CTTR cutthroat trout, DV Dolly Varden, JUVCN juvenile centrarchid, LM lamprey, LND longnose dace, MIN minnows, PM Northern pike minnow, RBTR rainbow trout, SLSU Salish sucker, SC sculpins, TSSB threespine stickleback. Amphibians: BLFR bullfrog, GRFR greenfrog, NWS north-western salamander, PTFR Pacific tree frog, RLFR red-legged frog, RSN rough-skinned newt.

similar response to area, which was positively correlated with axis 2. In addition to significant correlations with environmental axes, several environmental variables were correlated. Wood was negatively correlated with aquatic vegetation ($r = -0{\cdot}63$) and positively correlated with organic matter ($r = 0{\cdot}57$). Maximum temperature and aquatic vegetation were positively correlated ($r = 0{\cdot}54$), as were maximum depth and variation in depth ($r = 0{\cdot}71$).

Discussion

The umbrella species concept was articulated more than 25 years ago; however, until recently, there have been few empirical studies evaluating the concept (Wilcox 1984; Caro 2010). The vast majority of studies that have been conducted have been in terrestrial systems in the context of reserve design (Roberge & Angelstam 2004). To our knowledge, this is the first test of the umbrella species approach where habitat has been specifically modified or created to benefit a potential umbrella species. Our study is also the first time the relationships between abundance and biomass of presumptive umbrella and co-occurring species and specific environmental attributes were tested. The investigation of the potential use of the umbrella species approach to guide and streamline restoration work is particularly important for freshwater and riparian ecosystems, which have suffered disproportionately from anthropogenic habitat degradation and modification (Sala *et al.* 2000).

Typically, assessments of the umbrella species approach involve comparing species richness, and less often abundance, of species in areas with and without a presumed umbrella species (Branton & Richardson 2011). Sites are typically classified based on the presence or absence of the umbrella species (e.g. Caro *et al.* 2004). Had we used those standard measures of presence or absence of the umbrella species, and species richness of co-occurring species, we would have concluded that conservation efforts designed for coho provide little or no benefit to other vertebrates. However, by evaluating abundance and biomass, we were able to determine that where coho is most productive, so are several listed vertebrate species and fishes more generally.

Our study provides the first evidence of an umbrella species functioning in a restored aquatic system for sensitive vertebrates and fish. While coho was more effective as an umbrella species for other fish than for amphibians, benthic invertebrate species richness and productivity were lower in ponds where coho were more productive. However, the strong positive relationships between coho and listed species, which included fish and an amphibian, suggest that taxonomic similarity may not be as important to the effectiveness of an umbrella species as association with similar habitat features or shared resource requirements (e.g. see Seddon & Leech 2008). The negative relationship between coho and benthic invertebrate richness, abundance and biomass is consistent with the fact that, in general, benthic invertebrates clustered away from coho in ordinations with environmental attributes of ponds. We cannot preclude the possibility that benthic invertebrate species richness and productivity were depressed through direct relationships (e.g. predation, competition) with coho. Finally, we found that what is measured may influence how successful a conservation treatment is perceived to be, as the relative sensitivity of the variables we measured generally increased as follows: species richness < abundance ≤ biomass for listed species and fish. There was no such trend for benthic invertebrates, and we could not assess the relative sensitivity for amphibians as there were no significant relationships to assess. Our explicit evaluation of the relative sensitivity of different metrics that may be used to test the viability of the umbrella species approach, together with an assessment of habitat features that may be associated with more desired restoration outcomes, provides support for the use of more

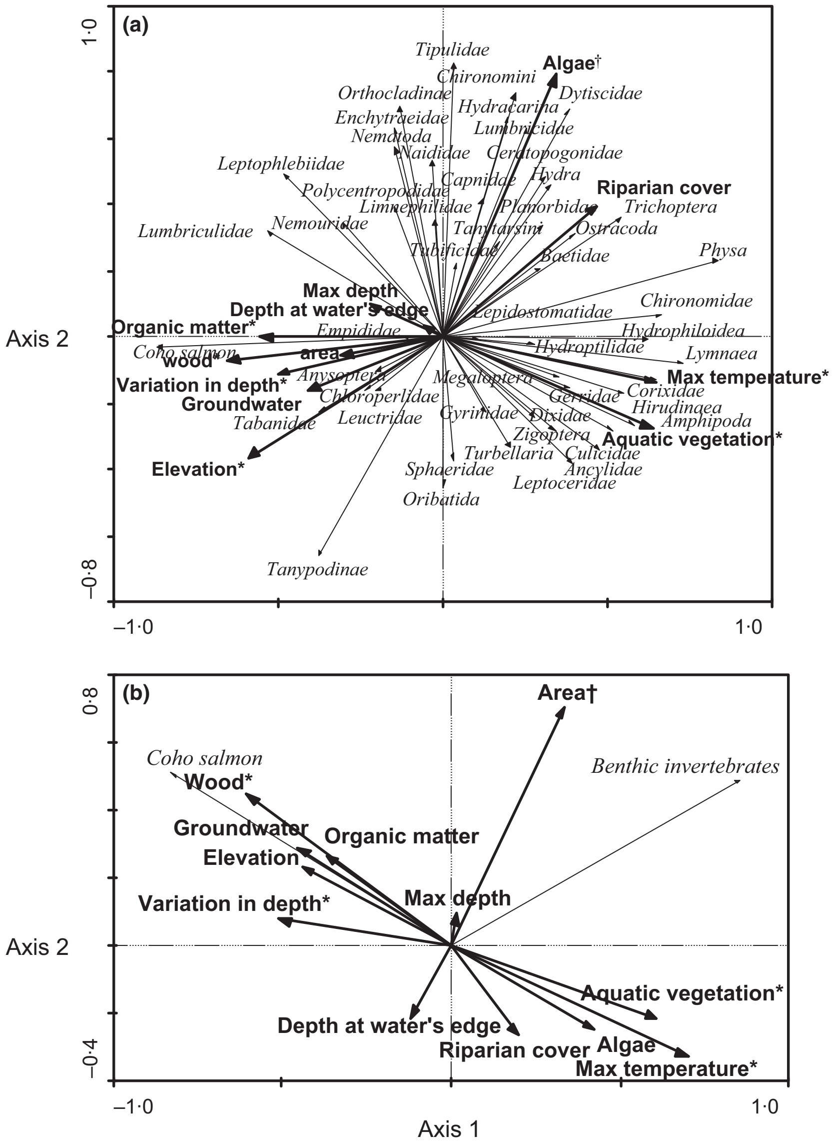

Fig. 4. Redundancy analysis ordination of the relative abundance (a) and biomass (b) of coho and benthic invertebrate species (small arrowheads) and environmental attributes (large arrowheads). Significant correlations marked *(Axis 1) and † (Axis 2). Axis 1: variation in depth and wood increase towards −1·0 and maximum temperature and aquatic vegetation increase towards 1·0. Axis 2: area increases towards 1·0.

labour-intensive metrics, such as abundance and biomass compared with species richness, at least in the validation phase of testing the umbrella species approach.

We found that the variation between watersheds within our study was an important contributor to overall variation that covered orders of magnitude, indicating inconsistent success of the restoration measures. Despite some differences among watersheds, the general trend that higher coho abundance and biomass were associated with higher abundance, biomass and species richness (coho abundance only) of listed species, and fish was robust across watersheds suggesting that our results are broadly applicable to this region.

The positive or non-significant relationships between coho abundance and biomass and the species richness, abundance and biomass of co-occurring vertebrate species are notable. Unlike reserve selection, exercises that aim to maximize species richness of co-occurring species, with restoration activities, there may be concerns that habitat restored for one species may result in unintended detrimental effects for co-occurring species (Roni *et al*. 2006; Suazo *et al*. 2009). In our study for vertebrates, we found either a positive (listed species, fish) or neutral (amphibians) relationship with coho, which is consistent with Roni (2003) who found the placement of large wood as a stream restoration treatment to be largely neutral for non-salmonid fish and an amphibian. Our results suggest that co-occurring vertebrates generally benefitted from pond restoration by virtue of their presence in the ponds, but did not, in the case of amphibians, parallel the response of coho.

The species composition of the community that populates restored habitat must also be considered when assessing the success of restoration. We found two exotic amphibians, bull frog *Lithobates catesbeianus* (Shaw, 1802) and green frog *L. clamitans* (Latreille, 1801), in several ponds with low coho productivity. This information is useful from a management perspective as those ponds were providing little benefit for coho, the target of the restoration, and were providing habitat for exotic species that may have a detrimental impact on native species. It is not clear whether the low coho productivity in those ponds was due to unfavourable conditions (e.g. extreme water temperatures), increased predation from the exotic species or other unmeasured factors.

Although all the ponds we studied were restored for coho, the environmental attributes of the ponds varied widely. Environmental variables explained significant variation in both vertebrate and invertebrate species abundance and biomass. Such as coho, the abundance and biomass of the listed species (cutthroat trout, Dolly Varden and northern red-legged frog) were associated with relatively higher elevations, more wood and less vegetation as aquatic structure and lower maximum temperatures. Increased biomass was also associated with groundwater fed ponds. It is not clear why the abundance and biomass of coho and listed species were associated with higher elevations; however, it is notable that our higher elevation ponds tended to be located in forests and lower elevation ponds tended to be in proximity of agriculture and low density residential areas. The lack of a significant correlation with pond area (mean = 4116 m^2, SE 903) is consistent with an assessment of smolt density as a function of pond area that found the optimal pond area threshold to be below 5000–10 000 m^3 with decreases in smolt density in larger ponds (Rosenfeld *et al.* 2008).

The association of coho and listed species with wood is consistent with the literature, which reports wood to be important for salmonids in both stream and pond environments (Bustard & Narver 1975; Roni & Quinn 2001; Giannico & Hinch 2003). It is not clear whether the abundance and biomass of coho and the listed species was higher where there was less aquatic vegetation due to the vegetation itself or due to the lack of wood in those ponds. Our results, which indicated higher biomass of coho and listed species in groundwater fed ponds and in ponds with lower maximum temperatures, are consistent with the literature that reports higher coho density in groundwater fed side channels with lower water temperatures in summer and warmer temperatures in winter (Morley *et al.* 2005).

The evaluation of habitat as a mechanism by which umbrella species can benefit co-occurring species provides conservation practitioners valuable insights regardless of whether or not the umbrella species is effective (e.g. Suter, Graf & Hess 2002; Ozaki *et al.* 2006). Primary design considerations should remain focused on the species that the restoration was motivated by. However, other features could be included as part of restoration projects broadening the overall conservation benefits if their inclusion would augment the restoration to provide benefits to co-occurring species, be neutral at worst for the target species and be feasible from an engineering and cost perspective (e.g. see Koper & Schmiegelow 2007). Equally, if there are habitat features associated with exotic species, such as the association of bull frog and green frog with warmer maximum water temperatures in our study, that are detrimental to native species, restored habitat could be designed to be less hospitable to those species.

Management implications

The resources are rarely made available to monitor the success of restoration projects and to determine whether the intended beneficiary of the project responded as planned. Our study provides evidence that a detailed evaluation of restoration projects, even years after they were implemented, delivers valuable information regarding what habitat features are associated with the intended response, in this case increased coho production, or unintended response, increased abundance of valued co-occurring species or nuisance exotic species. The resolution we gained by considering the relative abundance of the umbrella and the co-occurring species allowed us to better assess the function of coho as an umbrella species as well as to identify habitat features of the restored ponds that were associated with higher productivity of both the umbrella and other species of interest.

In the future, it would be beneficial for the results of our study to be tested in other parts of the Pacific Northwest where they could contribute to regional conservation and restoration planning. In addition, this approach, which incorporates evaluating multiple measures of umbrella and co-occurring species and environmental features of restored habitat, should be tested pre- and post-restoration and to monitor recovery through time. An investment in targeted, intensive monitoring, including the explicit evaluation of habitat features, to finetune restoration strategies can ensure both umbrella and co-occurring species can benefit when used as part of a regional conservation plan and make future restoration both more efficient and successful (Lambeck 1997; Lindenmayer *et al.* 2002).

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Data accessibility

Pond Characteristics, Vertebrate Mass, Invertebrate Mass and Invertebrate Abundance: Data available from the Dryad Digital Repository: http://doi.org/10.5061/dryad(Branton & Richardson 2014).

References

Anonymous (1995) *Lake and Stream Inventory Standards and Procedures*. British Columbia Ministry of Environment, Lands and Parks, Fisheries Branch, Victoria.

Barbour, M.T., Gerritsen, J., Snyder, B.D. & Stribling, J.B. (1999) *Rapid Bioassessment Protocols for Use in Streams and Wadeable Rivers: Periphyton, Benthic Macroinvertebrates and Fish*, 2nd edn. EPA 841-B-99-002. U.S.E.P.A.O., Washington DC.

Beechie, T., Beamer, E. & Wasserman, L. (1994) Estimating coho salmon rearing habitat and smolt production losses in a large river basin, and implications for restoration. *North American Journal of Fisheries Management*, **14**, 797–811.

Berger, J. (1997) Population constraints associated with the use of black rhinos as an umbrella species for desert herbivores. *Conservation Biology*, **11**, 69–78.

Betrus, C.J., Fleishman, E. & Blair, R.B. (2005) Cross-taxonomic potential and spatial transferability of an umbrella species index. *Journal of Environmental Management*, **74**, 79–87.

ter Braak, C.J.F. & Smilauer, P. (2002) *CANOCO Reference Manual and CanoDraw for Windows User's Guide: Software for Canonical Community Ordination (version 4.5)*. Microcomputer Power, Ithaca, NY, USA.

Branton, M. & Richardson, J.S. (2011) Assessing the value of the umbrella-species concept for conservation planning with meta-analysis. *Conservation Biology*, **25**, 9–20.

Branton, M. & Richardson, J.S. (2014) Data from: A test of the umbrella species approach in restored floodplain ponds. *Dryad Digital Repository*: http://doi.org/10.5061.

Bryant, M.D. (2004) Evaluating stream habitat survey data and statistical power using an example from Southeast Alaska. *North American Journal of Fisheries Management*, **24**, 1353–1362.

Bustard, D.R. & Narver, D.W. (1975) Aspects of the winter ecology of juvenile coho salmon (*Oncorhynchus kisutch*) and steelhead (*Salmo gairneri*). *Journal of Fisheries Research Board of Canada*, **32**, 667–680.

Caro, T.M. (2003) Umbrella species: critique and lessons from East Africa. *Animal Conservation*, **6**, 171–181.

Caro, T. (2010) *Conservation by Proxy: Indicatore, Umbrella, Keystone, Flagship and Other Surrogate Species*. Island Press, Washington, DC.

Caro, T.M. & O'Doherty, G. (1999) On the use of surrogate species in conservation biology. *Conservation Biology*, **13**, 805–814.

Caro, T., Engilis, A., Fitzherbert, E. & Gardner, T. (2004) Preliminary assessment of the flagship species concept at a small scale. *Animal Conservation*, **7**, 63–70.

Colwell, R.K. (2009) EstimateS, Version 8.2: Statistical estimation of species richness and shared species from samples (Software and users guide). Freeware for Windows and Mac OS.

Corkran, C.C. & Thoms, C. (2006) *Amphibians of Oregon, Washington and British Columbia*, 2nd edn. Lone Pine, Vancouver, British Columbia.

Fleishman, E., Murphy, D.D. & Brussard, P.F. (2000) A new method for selection of umbrella species for conservation planning. *Ecological Applications*, **10**, 569–579.

Fleishman, E., Noss, R.F. & Noon, B.R. (2006) Utility and limitations of species richness metrics for conservation planning. *Ecological Indicators*, **6**, 543.

Giannico, G.R. & Hinch, S.G. (2003) The effect of wood and temperature on juvenile coho salmon winter movement, growth, density and survival in side-channels. *River Research and Applications*, **19**, 219–231.

Johnston, N.T. & Slaney, P.A. (1996). *Fish Habitat Assessment Procedures*. Watershed Restoration Technical Circular Number 8. (ed. W.R. Program), pp. 97. Ministry of Environment, Lands and Parks and Ministry of Forests, British Columbia.

Koper, N. & Schmiegelow, F.K.A. (2007) Does management for duck productivity affect songbird nesting success? *Journal of Wildlife Management*, **71**, 2249–2257.

Lambeck, R.J. (1997) Focal species: a multi-species umbrella for nature conservation. *Conservation Biology*, **11**, 849–856.

Lindenmayer, D.B., Manning, A.D., Smith, P.L., Possingham, H.P., Fischer, J., Oliver, I. & McCarthy, M.A. (2002) The focal-species approach and landscape restoration: a critique. *Conservation Biology*, **16**, 338–345.

Margules, C.R. & Pressey, R.L. (2000) Systematic conservation planning. *Nature*, **405**, 243–253.

Morley, S.A., Garcia, P.S., Bennett, T.R. & Roni, P. (2005) Juvenile salmonid (*Oncorhynchus spp.*) use of constructed and natural side channels in Pacific Northwest rivers. *Canadian Journal of Fisheries and Aquatic Science*, **62**, 2811–2821.

Olson, D.H., Leonard, W.P. & Bury, R.B. (1997) *Sampling Amphibians in Lentic Habitats: Methods and Approaches for the Pacific Northwest*. Northwest Fauna Number 4. Society for Northwestern Vertebrate Biology, Olympia, Washington.

Ozaki, K., Isono, M., Kawahara, T., Iida, S., Kudo, T. & Fukuyama, K. (2006) A mechanistic approach to evaluation of umbrella species as conservation surrogates. *Conservation Biology*, **20**, 1507–1515.

Pacific Fisheries Resource Conservation Council (2010) Annual Report 2009, Pacific Fisheries Resource Conservation Council.

Pakkala, T., Pellikka, J. & Linden, H. (2003) Capercaillie *Tetrao urogallus* - a good candidate for an umbrella species in taiga forests. *Wildlife Biology*, **9**, 309–316.

Peterman, R.M. (1990) Statistical power analysis can improve fisheries research and management. *Canadian Journal of Fisheries and Aquatic Sciences*, **47**, 2–15.

Recovery Team for Salish Sucker (2010) *Recovery Strategy for the Salish Sucker* (Catostomus *sp.*) *in Canada* [Draft]. Species at Risk Act Recovery Strategy Series, Fisheries and Oceans Canada, Vancouver.

Roberge, J.M. & Angelstam, P.E.R. (2004) Usefulness of the umbrella species concept as a conservation tool. *Conservation Biology*, **18**, 76–85.

Roni, P. (2003) Responses of benthic fishes and giant salamanders to placement of large woody debris in small Pacific Northwest streams. *North American Journal of Fisheries Management*, **23**, 1087–1097.

Roni, P. (2005) Overview and background. *Monitoring Stream and Watershed Restoration* (ed. P. Roni), pp. 1–11. American Fisheries Society, Bethseda, Maryland, USA.

Roni, P. & Quinn, T.P. (2001) Density and size of juvenile salmonids in response to placement of large woody debris in western Oregon and Washington streams. *Canadian Journal of Fisheries and Aquatic Science*, **58**, 282–292.

Roni, P., Bennett, T., Morley, S., Pess, G.R., Hanson, K., Slyke, D.V. & Olmstead, P. (2006) Rehabilitation of bedrock stream channels: the effects of boulder weir placement on aquatic habitat and biota. *River Research and Applications*, **22**, 967–980.

Rosenfeld, J.S., Raeburn, E., Carrier, P.C. & Johnson, R. (2008) Effects of side channel structure on productivity of floodplain habitats for juvenile coho salmon. *North American Journal of Fisheries Management*, **28**, 1108–1119.

Sala, O.E., Chapin, F.S. III, Armesto, J.J., Berlow, E., Bloomfield, J., Dirzo, R. *et al.* (2000) Global Biodiversity Scenarios for the Year 2100. *Science*, **287**, 1770–1774.

Sandercock, F.K. (1991) Life history of coho salmon (*Oncorhynchus kisutch*). *Pacific Salmon Life Histories* (eds C. Groot & L. Margolis), pp. 395–445. UBC Press, Vancouver.

Seddon, P.J. & Leech, T. (2008) Conservation short cut, or long and winding road? A critique of umbrella species criteria. *Oryx*, **42**, 240–245.

Suazo, A.A., Fauth, J.E., Roth, J.D., Parkinson, C.L. & Stout, I.J. (2009) Responses of small rodents to habitat restoration and management for the imperiled Florida Scrub-Jay. *Biological Conservation*, **142**, 2322–2328.

Suter, W., Graf, R.F. & Hess, R. (2002) Capercaillie (*Tetrao urogallus*) and avian biodiversity: testing the umbrella-species concept. *Conservation Biology*, **16**, 778–788.

Vitousek, P.M., Mooney, H.A., Lubchenco, J. & Melillo, J.M. (1997) Human domination of earth's ecosystems. *Science*, New Series, **277**, 494–499.

Wilcox, B.A. (1984) *In situ* conservation of genetic resources: determinants of minimum area requirements. *National Parks, Conservation and Development: The Role of Protected Areas in Sustaining Society* (eds M.J.A. & K.R. Miller), pp. 639–647. Smithsonian Institution Press, Washington, D.C.

Wissinger, S.A., Greig, H. & McIntosh, A. (2009) Absence of species replacements between permanent and temporary lentic communities in New Zealand. *Journal of North American Benthological Society*, **28**, 12–23.

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Supporting Information

Additional Supporting Information may be found in the online version of this article.

Table S1. Summary of restoration techniques and pond characteristics.

Table S2. Full species list.